

Timothy Reuter

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Professional Experience:

Corys | Software Engineer - Transportation | Dec 2024 - Present | C++

- Worked with C++ in a custom engine consisting of 50+ modules to create locomotive simulations.
- Implemented behaviors for express wireless railroad crossings, giving the train operators the options of how they want specific railroad crossings to behave based on their route's planned stops.
- Designed and implemented speed enforcement logic driven by rail signal frequencies and rules to accurately replicate real world train safety systems.
- Debugged and fixed bugs related to the student scoring system to improve accuracy of the grading.

SSE Inc | Boeing Contract Software Engineer I | Oct 2021 - Oct 2024 | Unity3D, C#, C++

- Contributed to five shipped simulation projects and a core framework, utilizing C++ and C#.
- Implemented real-world functionality for aircrafts in Unity/RHEL(C++) for PC/VR/AR.
- Collaborated with system engineers and artists in an Agile environment to implement and refine complex functionality.
- Led development of a real-time procedure creation framework, enabling simulations to reflect changes immediately, creating a flexible system for customers.
- Develop Unity editor tools to streamline art asset implementation, build processes, project cleanup, and VR to AR conversions, eliminating common build failures and reducing procedure implementation time by 20%.
- Debugged and optimized core system code to enhance performance and consistency in the military simulations

SSE Inc | Unity Programmer | Oct 2018 - Jul 2020 | Unity3D, C#, C++, ADA

- Designed and Developed a real-time maintenance grading system, enabling instructors to track student performance and safety compliance in a networked environment, similar to a quest system found in games.
 - Contained a comprehensive tracking system to monitor student actions, providing instructors with actionable insights into success, failure, and missed safety measures.
- Served as merge master, performing thorough final code reviews and checks to ensure seamless integration of work into projects, upholding code quality and functionality.
- Primary point of contact for multiple systems, providing support and training to team members to improve onboarding and their contributions to the projects.

Freelance | 3D ArchViz Artist | Jan 2016 to Jun 2018 | Blender, Substance Designer

- Created high-quality, photorealistic 3D models and renders of buildings for use in ArchViz projects.

Magic Spells Studios | Asset Developer | Aug 2015 to Dec 2015 | Unity3D, ZBrush, Blender

- Create PBR characters/props from concept to finish for Hack, Slash and Backstab (PC/Xbox One - [Steam page](#)).

Projects:

Starlit Horizon

- Implemented a flexible weapon system in Unreal using the Gameplay Ability System, C++, and Blueprints.
- Allows for seamless mid-combat weapon swapping, continuing flow of attack sequences between each weapon.

Technical Skills and Software:

Programming Languages: C# | C++ | Python | Java | SQL | Javascript

Game Engine/Frameworks: Unreal Engine (Gameplay Ability System, C++, Blueprints), Unity3D

Software & Tools: Visual Studio, Git, Perforce, Blender, ZBrush, Substance Designer/Painter, Maya

Mathematics: Gameplay Systems, Linear Algebra, Transform Matrices, Collision Detection, Quaternions

Education:

Rochester Institute of Technology:

B.S. Game Design and Development - Minor: Japanese Language and Culture - GPA: 3.8